

Test Case 1: (Same as textbook examples 1.8.1 & 1.8.3)

ω_{opt} , Ω_{opt} for C_{Ld} , $(\alpha - \alpha_{L0})$ for C_L , $\delta_a = 5^\circ$, $p^- = 0$

Input File:

```
{
  "wing": {
    "planform": {
      "aspect_ratio": 8.0,
      "taper_ratio": 1.0
    },
    "airfoil_lift_slope": 6.283185307179590,
    "nodes_per_semispan": 49,
    "washout": {
      "distribution": "optimum",
      "distribution_options": [
        "none: no twist",
        "linear: linear twist distribution",
        "optimum: optimum twist distribution"
      ],
      "amount[deg]": "optimum",
      "amount_options": [
        "value: real number in degrees",
        "optimum: requires CL_design"
      ],
      "CL_design": 0.4
    },
    "aileron": {
      "begin[z/b]": 0.25,
      "end[z/b]": 0.45,
      "begin[cf/c]": 0.18,
      "end[cf/c]": 0.18,
      "hinge_efficiency": 0.85
    }
  },
  "condition": {
    "alpha_root[deg]": "CL",
    "alpha_options": [
      "value: real number in degrees",
      "CL: requires CL"
    ],
    "CL": 0.4,
    "aileron_deflection[deg]": 5,
    "pbar": 0.0,
    "pbar_options": [
      "value: real number nondimensional rolling rate",
      "steady: computes steady-state rolling rate"
    ]
  },
  "view": {
    "planform": true,
    "washout_distribution": true,
    "aileron_distribution": true,
    "CL_hat_distributions": true,
    "CL_tilde_distributions": true
  }
}
```

Display Results:

-----Kappa Results-----

kL: 0.039036

CLA: 4.837704

eW: 0.176278

kD: 0.067611

kDL: 0.137937

kDW: 0.070353

Cl,da: -0.242103

Cl,p: -0.591356

---Operating Results---

CL: 0.400000

CDi: 0.007183

Cl: -0.021127

Cn: 0.001009

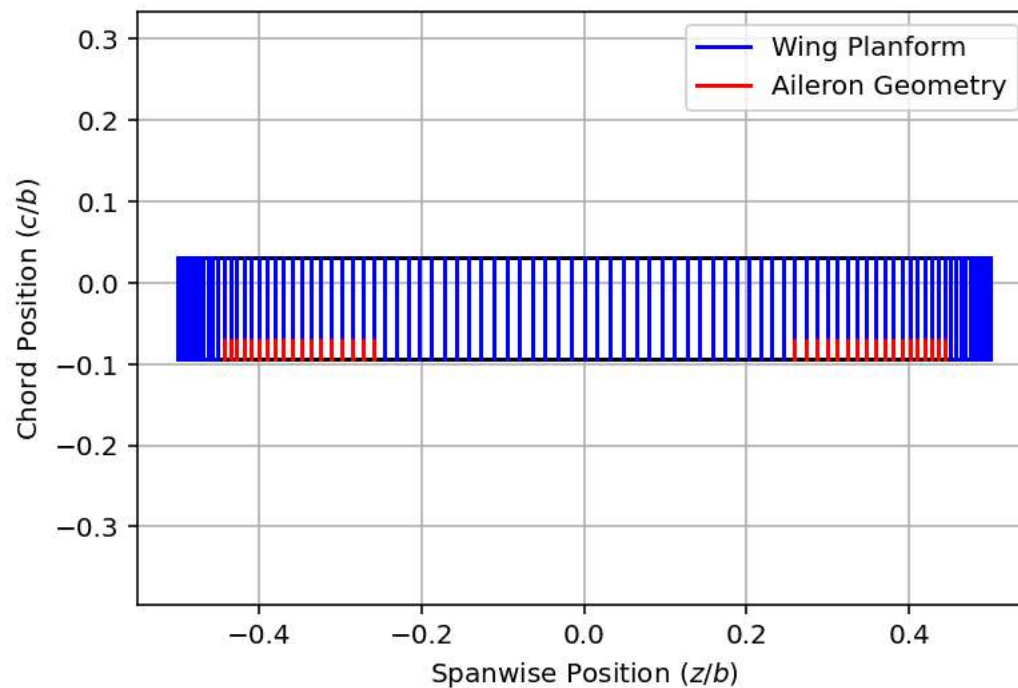
p: 0.000000

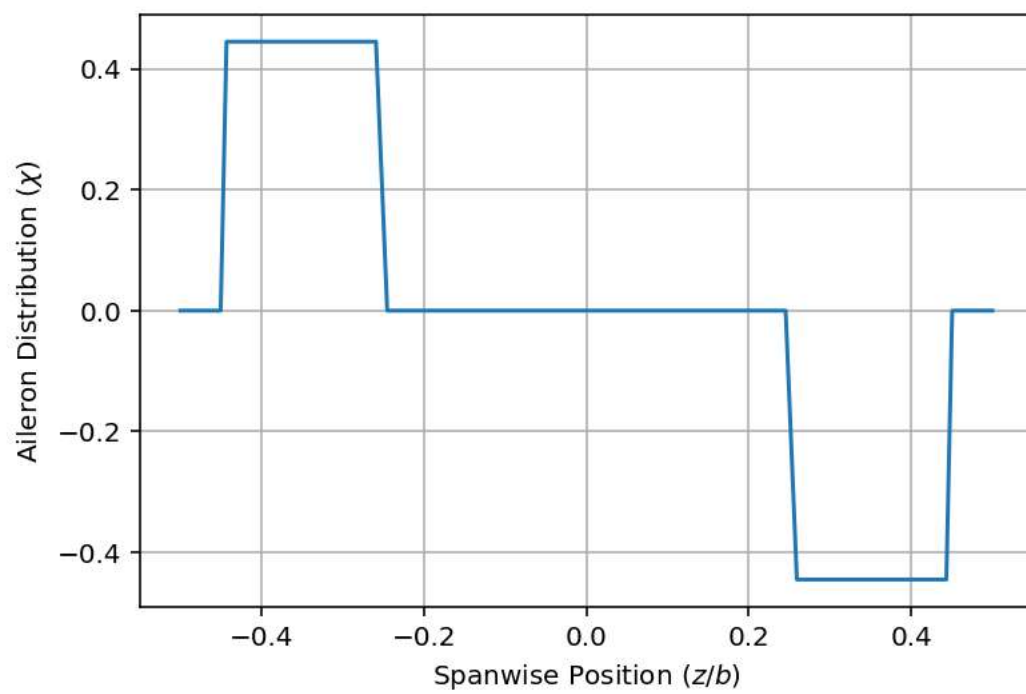
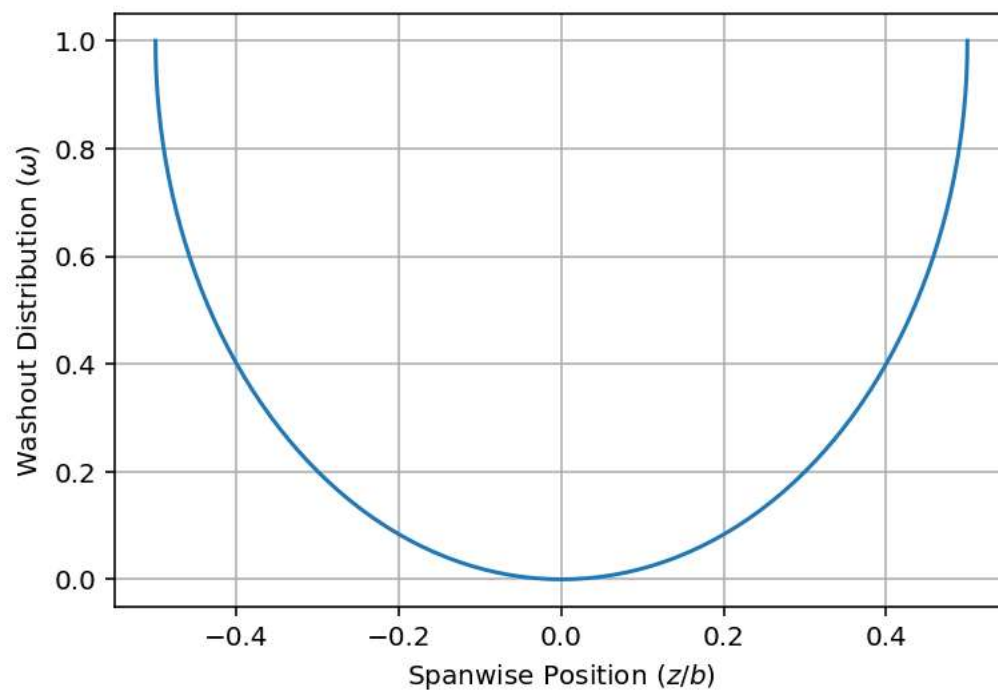
Select Formatted Text File Results: (C , C^{-1} , a_n , b_n , c_n , and d_n)

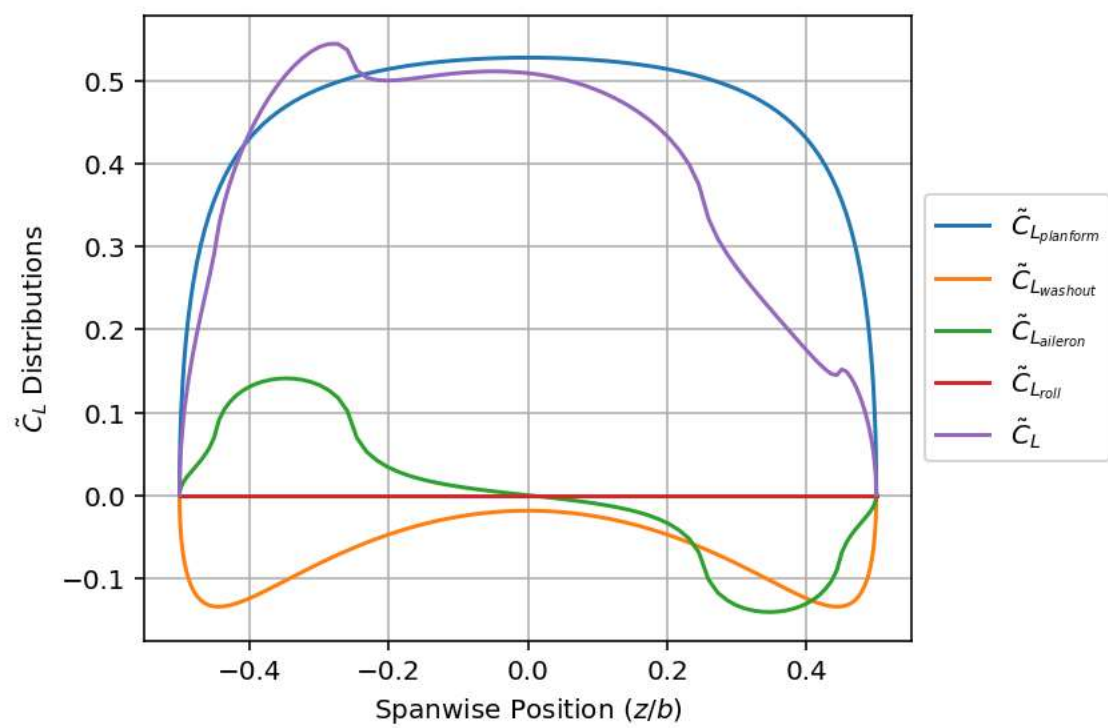
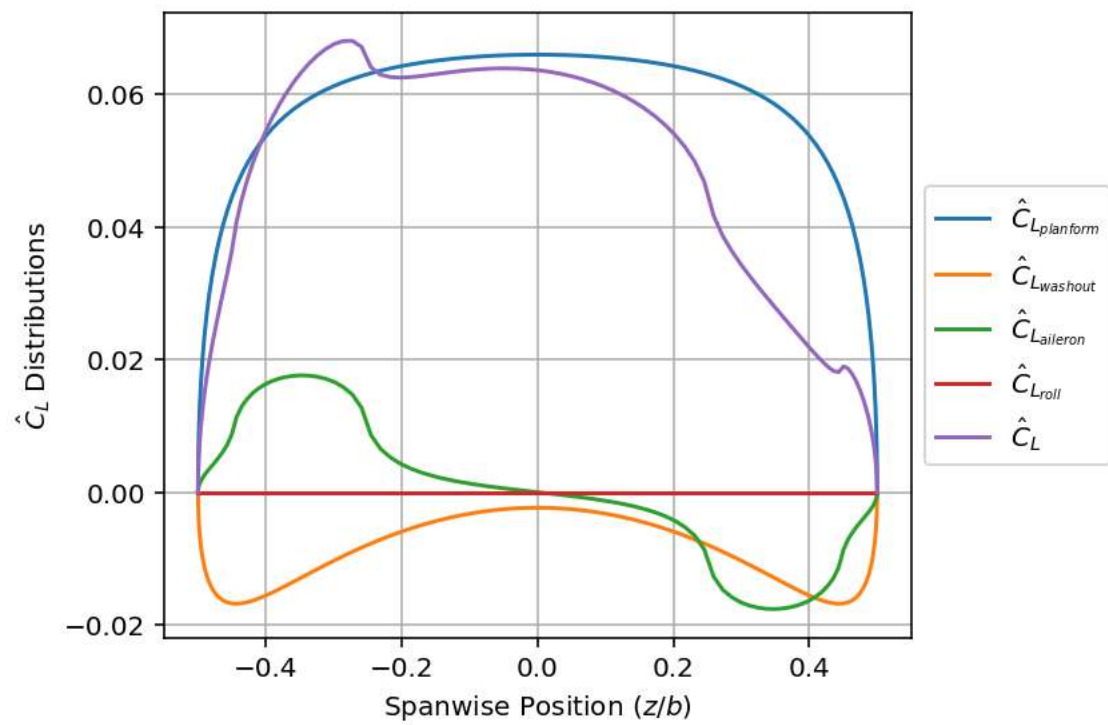
Fourier Coefficients (Columns):

n	a_n	b_n	c_n	d_n
1	0.192486	0.033931	-0.000000	0.000000
2	0.000000	0.000000	0.038532	0.094117
3	0.027408	0.032789	-0.000000	-0.000000
4	0.000000	0.000000	0.003351	0.013261
.				
.				
.				
98	0.000000	0.000000	-0.000018	0.000002
99	0.000001	0.000002	-0.000000	0.000000

Plot Results:







Test Case 2:

$\omega_{\text{linear}}, \Omega = 2.0^\circ, (\alpha - \alpha_{L0}) = 4.0^\circ, \delta_a = -1^\circ, p_{ss}^-$

Input File:

```
{
  "wing": {
    "planform": {
      "aspect_ratio": 10.0,
      "taper_ratio": 0.5
    },
    "airfoil_lift_slope": 6.283185307179590,
    "nodes_per_semispan": 49,
    "washout": {
      "distribution": "linear",
      "distribution_options": [
        "none: no twist",
        "linear: linear twist distribution",
        "optimum: optimum twist distribution"
      ],
      "amount[deg]": 2.0,
      "amount_options": [
        "value: real number in degrees",
        "optimum: requires CL_design"
      ],
      "CL_design": 0.45
    },
    "aileron": {
      "begin[z/b]": 0.30,
      "end[z/b]": 0.45,
      "begin[cf/c]": 0.1,
      "end[cf/c]": 0.2,
      "hinge_efficiency": 0.75
    }
  },
  "condition": {
    "alpha_root[deg]": 4.0,
    "alpha_options": [
      "value: real number in degrees",
      "CL: requires CL"
    ],
    "CL": 0.1,
    "aileron_deflection[deg]": -1.0,
    "pbar": "steady",
    "pbar_options": [
      "value: real number nondimensional rolling rate",
      "steady: computes steady-state rolling rate"
    ]
  },
  "view": {
    "planform": true,
    "washout_distribution": true,
    "aileron_distribution": true,
    "CL_hat_distributions": true,
    "CL_tilde_distributions": true
  }
}
```

Display Results:

-----Kappa Results-----

kL: 0.012291

CLA: 5.172412

eW: 0.433276

kD: 0.022711

kDL: 0.051222

kDW: 0.139058

Cl,da: -0.141178

Cl,p: -0.585367

---Operating Results---

CL: 0.282874

CDi: 0.002672

Cl: 0.000000

Cn: -0.000129

p: 0.004209

Select Formatted Text File Results: (C , C^{-1} , a_n , b_n , c_n , and d_n)

Fourier Coefficients (Columns):

n	a_n	b_n	c_n	d_n
1	0.164643	0.071336	0.000000	0.000000
2	0.000000	0.000000	0.017975	0.074531
3	0.007841	0.037937	-0.000000	0.000000
4	-0.000000	-0.000000	0.005277	0.007215
.				
.				
.				
98	0.000000	0.000000	-0.000018	0.000004
99	0.000002	0.000003	-0.000000	0.000000

Plot Results:

